Mechatronic systems simulation [146029] - MOD2 — 2025, August 28^{th}

Surname		Name		Mat. number	

Signature ____________________

1

Consider the following DAE system.

$$
\begin{cases}
x'(t) = x(t)y(t) \\
x(t) = t \\
x(t)z(t) = 1
\end{cases}
$$

rewrite in the form

$$\boldsymbol{E}(\boldsymbol{x},t)\boldsymbol{x}' = \boldsymbol{G}(\boldsymbol{x},t)$$

perform index reduction

$$\boldsymbol{E}^{(1)}(\boldsymbol{x},t)\boldsymbol{x}' = \boldsymbol{G}^{(1)}(\boldsymbol{x},t), \qquad \cdots \qquad \boldsymbol{E}^{(k)}(\boldsymbol{x},t)\boldsymbol{x}' = \boldsymbol{G}^{(k)}(\boldsymbol{x},t),$$

until $\boldsymbol{E}^{(k)}(\boldsymbol{x},t)$ is non singular and DAE is reduced to an ODE. Compute:

Differential index:
List of all invariants (or algebraic constraints):

Matrices $\boldsymbol{E}^{(j)}(\boldsymbol{x},t)$ and vector $\boldsymbol{G}^{(j)}(\boldsymbol{x},t)$ at each step of index reduction

2

Consider the following DAE system:

$$
\begin{cases}
x''(t) = \lambda(t)y(t) - y'(t) \\
y''(t) = \lambda(t)x(t) \\
x(t)y(t) = 1 + t
\end{cases}
$$

reduce to order 1 and reduce the DAE to index 1 compute also invariants. (Suggestion use dummy variables)

Differential index:
DAE reduced to order 1
DAE reduced to index 1

List of all invariants (or algebraic constraints):

Projection step to $xy = 1 + t$ (fist invariant)